

Ryan Nguyen

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EDUCATION

University of Massachusetts Amherst, B.S. in Computer Science & Mathematics Aug 2022 - May 2026
Relevant Coursework: Data Structures & Algorithms, Web Programming, Computer Systems, Algorithm Design, Artificial Intelligence, Search Engines

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL
Frameworks & Tools: React, Node.js, Express.js, Django, Git, Docker, Figma, PyTorch, PostgreSQL, MongoDB, Sequelize, ESP32, Pinecone

WORK EXPERIENCES

University of Massachusetts Amherst **Amherst, MA**
Undergraduate Course Assistant | CS198C Teaching Assistant & CS Tutor Sep. 2025 - Present

- Hold **4-hour** weekly office hours for students, addressing questions and explaining core concepts in C programming and Unix/Linux systems.
- Regularly monitor and respond to student questions on the course forum, maintaining an average response time of under **2 hours**.
- Tutor **670+** students across CS160 (OOP), Advanced Python, and CS311 (Intro to Algorithms) as part of pedagogy practicum.
- Led workshops for **250+** students designed to facilitate student comprehension on complex topics in C programming.

Gamestop, Corp. **Braintree, MA**
Retail Keyholder & Seasonal Worker Jul. 2024 – Jun. 2025

- Processed **150** customer transactions per week, handling **\$3,500** in cash with zero discrepancies by following precise reconciliation procedures
- Audited inventory across **15** aisles each week, identifying and resolving discrepancies within **24 hours** to maintain accurate stock levels.

PROJECTS

RemoteRide | ESP32, C++, HTML, CSS, JavaScript Jan. 2025 – Jun. 2025

- Collaborated in a team of **3** to build a battery-powered toy car controlled via an ESP32 microcontroller with a custom web interface for real-time motor and servo control.
- Implemented live tracking of the car’s location through the web interface, allowing users to monitor and control the vehicle remotely.
- Integrated movement control based on finger position detected by onboard camera, enhancing user interactivity.

UKart | Node.js, Express.js, SQLite, Sequelize, CSS Sep. 2024 – Dec. 2024

- Led a team of **6** in building a full-stack marketplace platform for college students to buy, sell, and exchange goods securely and efficiently.
- Implemented a SQLite database with Sequelize ORM, ensuring efficient data storage and retrieval for listings and user profiles.
- Contributed to both frontend and backend, implementing key features including item listings, user profiles, and user verification.

F1 Nab | Python, CSS, React, Node.js, Pinecone, RAG, SQLite, Flask Nov. 2024 - Nov. 2024

- Led a team of **4** to build a web app that embeds F1 TV into a web app, allowing users to watch races and chat with the bot simultaneously.
- Developed an interactive chatbot to provide real-time insights into F1 races, helping new fans understand terminology, strategy calls, and race events.
- Implemented a backend pipeline combining race data from Ergast API, vector embeddings with Pinecone, and Gemini API integration to deliver fast responses.

Mafia | Python, Jupyter Notebook, OpenAI API Oct. 2024 - Dec. 2024

- Collaborated in a team of **3** to develop a real-time AI-driven Mafia/Werewolf game using OpenAI’s language models for smooth storytelling and player interactions.
- Designed and implemented RESTful API integrations with OpenAI endpoints to enable dynamic, context-aware AI responses for real-time gameplay.